



Does learning longer improve student achievement? Evidence from online education of graduating students in a high school during COVID-19 period

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ABSTRACT

This paper examines the effect of online learning time on graduating students' test scores in a senior high school. Decisions regarding online education, including those related to participation and learning hours, are endogenous due to both reverse causality and omitted variables. This paper is the result of the natural experiment of the outbreak of COVID-19, which made every student to participate in online education when the spring semester began. In addition, this paper uses a value-added model controlling for the scores that preceded online education, which is a sufficient statistic of students' unobserved ability and motivation. If this cannot completely eliminate the endogeneity problem, it should be able to largely alleviate the problem. The results indicate that: online education has positive but limited impacts on test scores on average, particularly those in the subject of math within the natural sciences track; top-tier students are most positively affected by online education; and the benefits of online education vary among students with different backgrounds. The quantile regression suggests that a 10% increase in online education time raises math test scores by more than 0.25 for the students between the 0.60th and 0.80th quantiles. Surprisingly, it is evident that online learning time has a significant negative effect for some students in certain subjects. Finally, online education neither widens nor narrows the inequality of students' test scores.

1. Introduction

The Internet allows new pedagogical modes to emerge, leading to enormous innovation and expansion during the recent decades. Traditional distance learning, which is mainly based on television and radio programs, provides opportunities to learn from home. However, the difficulty with these methods was that students were not able to interact with other students or ask professors questions, because everything was broadcasted to them. But online distance learning (ODL) overcame this difficulty and improved interactivity for learners, instead of being taught passively (Sumner, 2000). In the beginning, online education was just a supplement to formal

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education. One typical example of online distance learning (ODL) is CALCampus (Computer Assisted Learning Center Campus). In 1995, it began to offer an online school to deliver instruction, learning materials and administration entirely via the Internet, so students and teachers could interact in real time.¹ Currently, online education has become an indispensable part of university life (Larreamendy-Joerns & Leinhardt, 2006). In 1993, Jones International University became the first fully web-based university in the US and offered master degree initially.² Since then, more and more universities have begun to implement completely online degree programs. Nowadays, online degrees are available in 137 popular subjects, with 532 schools for bachelor's degree and 586 schools for master's degree in the US.³ Moreover, it is becoming more popular worldwide. The number of online education users in China has nearly doubled in three years, from 140 million in 2016 to 230 million in 2019.⁴ In the meantime, people have changed their attitude towards online education for at least three reasons. First, online students have access to courses at any time and at any place, which is more cost-efficient for attending class (Barbara, Yuki, Robert, & Jones, 2009). Second, the same instructional content may narrow the achievement gap resulting from uneven teaching quality. Third, since students can watch online courses repeatedly after school, online education time does not affect their normal learning trajectory (Loeb, 2020).

This paper examines the effects of online education time on students' achievement. Although learning time is one of the most critical inputs in the education production function (Brown & Saks, 1987; Coates, 2003) and online education is developing rapidly, there is very limited discussion on this topic in the literature. Previous studies in this field compare the effectiveness of offline and online education but do not arrive at a consensus. Some studies find that online education cannot be as effective as offline education (e.g., Brown & Liedholm, 2002; Heppen et al., 2017; Johnson, Aragon, Shaik, & Palma-Rivas, 2000), while others suggest that online education does not differ considerably from offline education (e.g., Allen, Bourhis, Burrell, & Mabry, 2002; Allen & Seaman, 2010; Cavanaugh, 2001; Machtmes & Asher, 2000; Neuhauser, 2002; Rovai, 2002). Mayer (2002) suggests that the observed difference between online and offline education is because of poor use of technology and incomplete analyses. In addition, some scholars have suggested that that online education can be superior to traditional education (e.g., Sitzmann, Kraiger, Stewart, et al., 2006).

Studies of learning time on students' achievement have a long history and mainly focus on two aspects: school instruction time and extracurricular learning time. Cooper, Charlton, Valentine, and Muhlenbruck (2000) find that all students can benefit from summer school, especially students from middle-class homes. In addition, after-school programs can strengthen students' noncognitive skills, such as responsibility and emotional learning (Gordon, Jacobs, & Wright, 2016; Ivy, Richards, Lawson, & Alameda-Lawson, 2018). Using instruction time to improve students' academic performance has been a policy issue. For instance, expanded learning time (ELT) is an emerging reform aimed at narrowing the achievement gap by adding hours to the traditional schedule. The assumption here is that more class time will result in higher achievement among disadvantaged students so that they can catch up with their peers. Intuitively, expanded instruction time can allow teachers to impart more knowledge and allow students to better understand lessons, but there is no consistent evidence for this in the literature. Some studies show that additional instruction time can increase all students' achievement (e.g., Cattaneo, Oggenfuss, & Wolter, 2016; Figlio, Holden, & Özek, 2018; Gijssels & Schmidt, 1995; Lavy, 2015; Nomi & Allensworth, 2013; Rivkin & Schiman, 2013). Others find no significant effect of expanded instruction time on the achievement of some students (e.g., Anderson, Ryan, & Shapiro, 1989; Baker, Fábrega, Galindo, & Mishook, 2004; Huebener, Kuger, & Marcus, 2017).

It should be noted that the learning time for online education differs from traditional instruction time. In previous studies, instruction time was mainly determined by the school curriculum, whereby there was no difference in instruction time for different students in the same class, and we are not able to tell how engaged students are during this time. Although students with different abilities and motivations have different "real learning times", they have the same instruction time statistically (Lavy, 2015; Rivkin & Schiman, 2013). Online education allows students to choose their learning time, especially when the supervision of teachers and parents is weak. Students can choose their "real learning time" based on their own judgment.

The main challenge in this field is the endogeneity problem.⁵ When online education is selective rather than compulsory, especially when it is complementary to offline education, some unobserved variables affect both online education participation decisions and schooling achievements. As a result, the usual OLS estimate is biased and inconsistent. Our context is in references to when, on December 31, 2019, the first COVID-19 infection case was reported in Wuhan, the capital city of Hubei Province, China (Kang et al., 2020). On January 23, 2020, the Chinese government locked down Wuhan, the epicenter of the pandemic. However, because the lockdown was just before the Chinese New Year when billions of people returned to their hometowns, the pandemic quickly spread across the whole country (Kang et al., 2020). As a direct result, all schools in China halted offline education and converted to fully online education after the Chinese New Year holiday. The sudden epidemic caused millions of teachers and students to enter online education. This provides us with a perfect natural experiment to identify the effectiveness of online education. Fortunately, we can

¹ See CALCampus website <http://www.calcampus.com/calc.htm>. It is a private, international, online distance learning institution, founded in 1982, which offers both college and high school courses through the Internet.

² See JIU website <https://www.jiu.edu.rs/about-JIU/index.html>, which is the pioneer in online learning.

³ See OnlineU website <https://www.onlineu.com/degrees>. It aims to provide accurate, detailed online degree information to inspire Americans to accomplish their education.

⁴ The data come from the 45th Statistical Report on China's Internet Development Status.

⁵ The other two challenges in this field are measurement error and comparability of students' achievement. Firstly, learning time of measurement error is unavoidable, which causes attenuation bias (Wooldridge, 2015). Secondly, whether the students' achievement is comparable is another concern. Usually, test content is not the same for all students. In our case, the results are completely comparable for students who are in the same track take the same exam.

avoid selection bias since epidemics make everyone participate in online education without a choice.

The other endogeneity problem may occur when students' ability and motivation affect both online learning time and academic performance. Since students can choose their learning time in online education, their ability and motivation would both affect decisions about time spent watching content online and academic performance, which results in another endogeneity problem. Our sample comes from graduating students in a high school. It is well known that the *Gaokao* (National College Entrance Examination) has a strong effect on students' futures, making graduating students highly motivated to achieve their goals (Liu & Wu, 2006). Students' ability and motivation are two key variables for their development, both of which are unobservable. However, because the lagged test score is a sufficient statistic for historical inputs and endowments (Todd & Wolpin, 2003), when we control for the students' initial academic performance – the test score exactly before the start of online education, this should be able to substantially alleviate the problem if it cannot completely overcome the endogeneity problem totally.

In this paper, we estimate the impact of online education on students' academic performance using data from 1141 high school students at Grade 12 who were about to attend the *Gaokao* in July 2020. Some scholars have studied similar topics and have discussed the relative questions of online education under COVID-19 (e.g., Bao, 2020; Kapasia et al., 2020; Thomas & Bryson, 2021). For example, Thomas and Bryson (2021) explore an inclusive approach to simultaneous teaching of proximate and online students. They are, however, focus on different problems from our research. This paper aims to answer three questions: (1) How effective is online education? (2) Does the effect differ among students, subjects, and personal characteristics? (3) Is the effect lasting? Our article also provides some guidelines for the development of online courses. At this stage, the efficacy of online courses is limited by equipment, teachers' training (Goodyear, Salmon, & Michael, 2001; Mayer, 2018), etc. The full realization of the benefits of online courses depends on technology, teachers, and class design. For example, professional training of teachers, online education equipment, online course curriculum setting, etc. will have a great impact on the effectiveness of online education. We still have a long way to go before the benefits of online courses are fully realized. We look forward to the future flourishing of online education.

The estimation results show that, on average, online education has significant but moderate effects on math test scores.⁶ When the online watching time for math increases by 10%, the score increases by approximately 0.344, holding other factors constant, but watching time has no statistically significant effect on other subjects in both natural sciences and social sciences.⁷ We also find that online education has a long-lasting effect on math for students among the natural sciences. In contrast, there is no persistent effect on other subjects.

The heterogeneity analysis results suggest that the top-tier students are most positively affected by online education in most subjects. For students with disparate backgrounds, there are also great differences in the benefits of online education. For instance, for students who use computers or tablets in online education, the coefficients are insignificant, implying that different tools may not affect online learning. The distributional effects show that students in the 0.60th to 0.80th quantiles of math can only increase their scores by more than 0.25 with a 10% increase in online watching time. However, for the liberal arts, we find that watching time has a significantly negative impact on student performance. These moderate distributional effects suggest that online education has no significant effect on test score inequalities.

The remainder of the paper is structured as follows. Section 2 describes our data and plots the raw correlation of the variables of interest. Section 3 presents our empirical model and estimation strategies. Section 4 presents the results, including the basic results, heterogeneity analyses, persistence of online education effects, distributional effects, and inequality effects of online education on test scores. Finally, Section 5 presents the conclusions of our research.

2. Data

2.1. Data sources and the background

This paper investigates academic performance of the graduating students of a senior high school in Shaanxi Province, China. When students finish their junior high school, they have two choices: academic high school and vocational high school. After vocational high school education, students can directly enter the labor market. *Gaokao*, which is held every summer in Mainland of China, takes place when students accomplished academic high-school education and are applying to 4-year universities or 3-year colleges, as Fig. 1 shows. Academic high schools have two tracks: natural sciences and social sciences. In general, students with better academic

⁶ The online course is a mixture of synchronous and asynchronous learning in our study. The school transferred all the content that should be granted to students in offline classrooms to the online platform, and arranged a timetable for students in different classes every day. Through DINGDING, an online live broadcast platform, students need to watch the live class at the specified time. Students could watch the videos again after the live class. Basically, students mainly attend live class. The time data are the sum of the time students watched initially and after the live class.

⁷ The length of online watching time firstly reflects the degree of knowledge and information students acquired, because it is an intuitive expression of whether the student is following the course. Secondly, it reflects the students' effort and self-discipline. Generally speaking, the students who are more hard working and more self-disciplined will have longer watch time, which is why we adopt the value-added model.

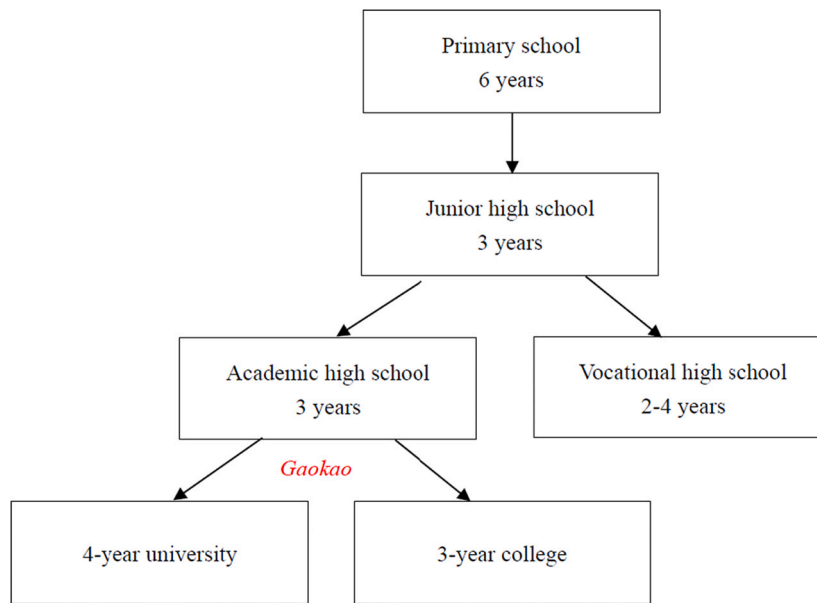


Fig. 1. The education system in China.

achievement prefer the natural sciences and vice versa.⁸ For the natural sciences track, the school classifies students into two top-tier, eight second-tier, and four third-tier classes based on students' previous academic performance in this high school, plus three returning student classes⁹; for the social sciences track, there are only two new and two returning student classes. The examination in both tracks includes Chinese, English, and math. The Chinese and English exams are the same between the two tracks, but math for the natural sciences track is often more difficult than that for the social sciences track. In addition, students in the natural sciences and social sciences are also tested on comprehensive science and comprehensive liberal arts, respectively. The comprehensive science test includes chemistry, physics, and biology, while the comprehensive liberal arts test includes history, politics, and geography. The data include not only students' test scores and online watching time but also a rich set of demographic characteristics, such as household consumption level. Our sample includes 1141 Grade 12 students in this high school who were about to attend the *Gaokao* in July 2020.

On February 8 the school shifted to fully online teaching. The course schedule was to take the same subject class every other day, for a total of 25 sessions. DINGDING is the only online teaching tool in the school and it recorded students' watching time information.¹⁰ Students could watch the videos again following the live class. The time data are the sum of the time students watched initially and after the live class. The students' online activity is to watch online courses, and the offline activity is to complete the homework assigned by teachers. There are indeed differences among subjects, mainly in the forms and intensity of offline homework. For example, Chinese homework focuses on reciting poems and math homework is mainly to complete exercises. Teachers do not require students to complete all of the homework in the whole process and do not inspect their quality, so it is hard to measure the offline education activities. After the online courses finished, students returned to school and had traditional classes. Before the epidemic, students took five exams in all, and they took five exams after the online education concluded. The last exam was the *Gaokao* (see Fig. 2 for the detailed timeline).

To obtain students' background information, we conducted a survey. The survey took place in July 2020. It consisted of 21 questions related to household characteristics, individual characteristics, learning status at home,¹¹ individual preference, and subjective evaluation, which may affect students' achievement (Hanushek, 1979). Finally, 1241 valid questionnaires were gathered.

⁸ The subjects in the first year of high school are exactly the same for everyone. At the beginning of the second year of high school, students need to choose a track, either natural sciences or social sciences. Generally speaking, students with better academic achievement in the first year of high school prefer natural sciences. It reflects parents' and students' preferences for they think it will be more competitive for natural sciences students in the labor market and it will be easier to find a job in the future, and this is why we look at the results of social sciences and natural sciences separately in the regression.

⁹ In the first four exams before the epidemic, the school reassigned the students in Classes 1 and 2 according to each score. They planned to classify the top 50 students into Class 1 and train them, but this plan was ultimately abandoned. We only kept those students who stayed in the same Class in our sample.

¹⁰ See details of DINGDING in Appendix D.

¹¹ For learning status at home, there are four variables to measure it. It includes students' daily learning hours, whether attend class on time, study status at home (ineffective, okay but not as good as school, or fulfilling), and attitude towards online courses (hate online courses, sleepy, absent-minded, or preoccupied). All are self-reported by students themselves.

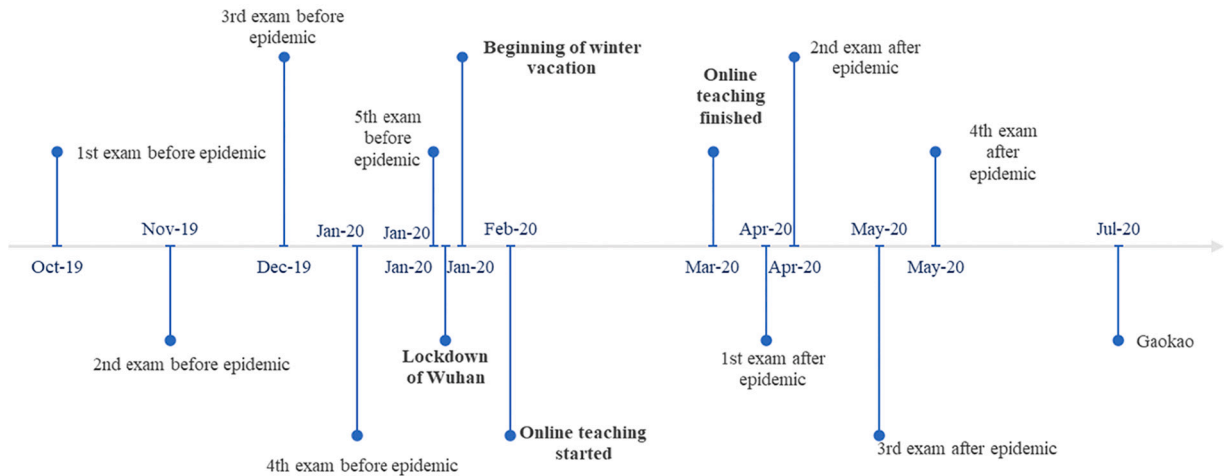


Fig. 2. Timeline of exams and online courses.

2.2. Data description

This paper uses grade-three student data for two tracks in one high school. Students graduated from high school in 2020. Each observation contains individual information (total online watching time by subject), test scores in 10 exams (including the *Gaokao*), and individual background information (household characteristics, individual characteristics, learning status at home, etc.). The sample used throughout our analysis consists of 1141 students, 934 in the natural sciences track distributed over 19 classes, and 207 in the social sciences track distributed over 4 classes.

2.3. Summary statistics

Table 1 presents summary statistics for natural sciences students and social sciences students. We focus on the 1st exam after online education concluded (hereafter 1st exam for short) and the *Gaokao*, given that the test score of the 1st exam reflects the purest effect of online education while controlling for the performance before the outbreak of the COVID-19, and the *Gaokao* is the most significant exam.¹² Table 1 shows the test scores of the above two exams, and presents the differences between scores for natural sciences and social sciences students. In general, social sciences students perform better than natural sciences students in Chinese and English.

On average, the parents' education is below the high school level, and the household consumption level is average or high. Social sciences students are particularly higher than natural sciences students for household consumption. Most students use cell phones rather than a computer or tablet to watch online courses, and parental supervision does not become more active during online education. For learning status at home, students' learning hours are less than 6 h per day in general. Only approximately one-third of the students attended online classes on time, and approximately 40% of the students were preoccupied when attending class. These statistics suggest students did not fully focus on their studies during online education. Finally, the majority of the students preferred offline education to online education, and students felt that the teaching effectiveness of math and comprehensive science worsened in the process of online education.

Our goal is to use student-level data to estimate the relationship between online watching time and test scores. In this section, we also generalize this approach by exploring the raw correlation between online watching time and the 1st exam score following online education, since this exam score was most directly affected by online education yet less affected by offline education after returning school. Fig. 3 plots the correlation results of natural sciences and social sciences students for Chinese, math, and English, as well as comprehensive science and liberal arts. For natural sciences students, there are positive relations for Chinese, English, and comprehensive science, but the correlation for math is negative. For social sciences students, the correlation is positive only for Chinese and English.

In addition, we plot raw correlation by various classifications among all classes. Appendix Figs. A.1 to A.4 present the relations between the log of online education time and the test score in the first test of each subject after the epidemic. For natural sciences track students, the longer the online education time, the higher the achievement obtained. Given the results shown in the figures in the appendix, we suggest that a performance gap in different classes does exist. In general, top-tier students perform best. The average grade of the second-tier students is similar to that of the returning students, while the third-tier students have the lowest average score. Although these descriptive correlations are insightful and foreshadow our results of the regression analysis, the other variables that could affect online watching time and students' achievements are not considered. In the next section, we use quantitative methods to analyze this problem.

¹² Online education concluded on 29th March 2020. The time of the 1st exam was 11th April 2020.

Table 1
Descriptive statistics.

Vairables	Natural sciences (N = 934)		Social sciences (N = 207)	
	Mean	Sd	Mean	Sd
Test score of the 1st exam after online education ^a				
Chinese score	101.861	10.108	105.536	6.253
English score	83.271	22.978	86.684	19.450
Math score (natural science)	79.085	20.546	–	–
Math score (social science)	–	–	67.879	19.166
Comprehensive science score	126.696	34.606	–	–
Comprehensive liberal arts score	–	–	210.063	21.667
Test score of Gaokao				
Chinese score	96.597	8.394	100.394	7.667
English score	82.772	42.817	82.267	16.088
Math score (natural science)	75.907	18.878	–	–
Math score (social science)	–	–	79.828	17.325
Comprehensive science score	163.257	33.980	–	–
Comprehensive liberal arts score	–	–	201.178	18.152
Household characteristics				
Father's education (high school and above = 1)	0.433	0.496	0.494	0.502
Mother's education (high school and above = 1)	0.384	0.487	0.413	0.494
Household consumption level (average or high = 1)	0.689	0.463	0.759	0.429
Parental supervision (more active = 1)	0.392	0.489	0.352	0.479
Device (computer or Tablet PC = 1)	0.262	0.440	0.296	0.458
Individual characteristics				
Gender (male = 1)	0.526	0.500	0.315	0.466
Urban (urban = 1)	0.222	0.416	0.272	0.446
Learning status at home				
Daily learning hours (more than 6 h = 1)	0.378	0.485	0.426	0.496
Whether attend class on time (always on time = 1)	0.352	0.478	0.358	0.481
Study status at home (okay or fulfilling = 1)	0.773	0.419	0.864	0.344
Attitude towards online courses (preoccupied = 1)	0.409	0.492	0.420	0.495
Individual preference				
Prefer online or traditional (indifferent or prefer online = 1)	0.239	0.427	0.235	0.425
Subjective evaluation				
Teaching effect of Chinese (indifferent or better = 1)	0.555	0.497	0.568	0.497
Teaching effect of Math (indifferent or better = 1)	0.345	0.476	0.370	0.484
Teaching effect of English (indifferent or better = 1)	0.505	0.500	0.481	0.501
Teaching effect of Comp. sci./lib. Arts (indifferent or better = 1)	0.306	0.461	0.519	0.501

Note: Variables of household characteristics, individual characteristics, learning status at home, individual preference, and subjective evaluation are dummy variables.

^a Chinese, Math, and English exams are worth 150 points each. Comprehensive science and Comprehensive liberal arts exams worth 300 points each.

3. Methodology

We adopt the value-added model in the education production function literature and estimate the following equation using student-level data:

$$Y_{ic} = \alpha Lntime_{ic} + \beta Y_{ic}^0 + \delta' X_{ic} + \gamma_c + \varepsilon_{ic} \quad (1)$$

where i denotes students, c denotes class; $Lntime_{ic}$ is the total watching time of student i in class c ; Y_{ic} is the raw scores of each exam after online education, while the term Y_{ic}^0 is the average score of 5 exams taken before online education; X_{ic} is a vector of control variables including gender, urban, parental education, household consumption level, parental supervision, watching device, everyday learning time, whether attend class on time, attitude in online courses, prefer online or traditional education, study status at home, and teaching effect of each subject; γ_c is the class fixed effect that controls for the potential confounding factors; ε_{ic} is the error term.

Students' motivation and ability are critical factors in their learning strategies (Boekaerts, 2011; Efklidesa, 2011), especially within computer-based learning environments (Duffy & Azevedo, 2015). Omitting these two vital factors can result in endogeneity problems, while the lagged test score can be viewed as a sufficient statistic for historical inputs and endowments (Todd & Wolpin, 2003). Thus, although we have no access to all past and present school and family input data, controlling for the initial scores from just before online education began can substantially alleviate the endogeneity problem when the problem cannot be completely overcome through other means.

Finally, standard errors are clustered at the class level throughout.

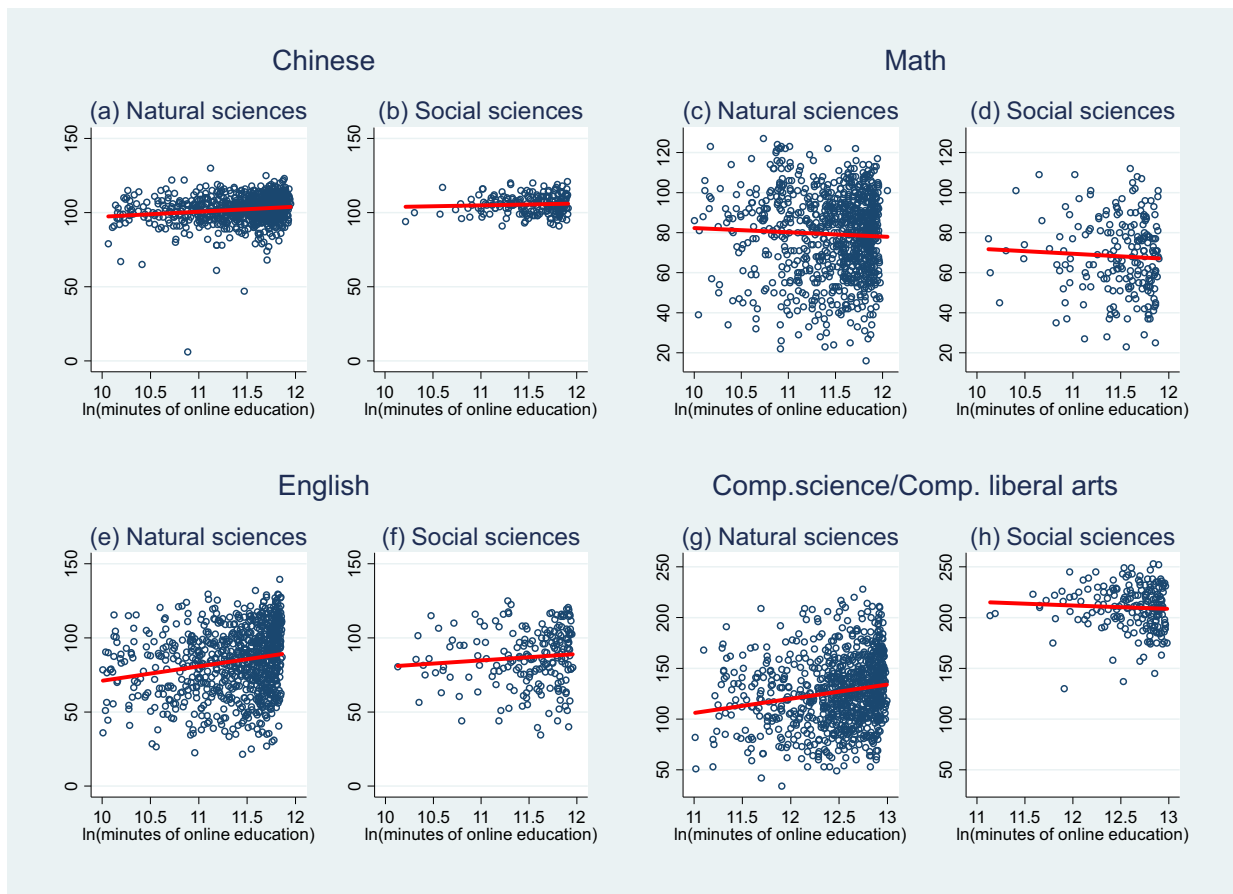


Fig. 3. The raw correlation between online watching time and the 1st exam score after online education.

4. Results

4.1. Baseline results

Table 2 presents the results of eq. (1) for the test scores of the natural sciences and social sciences tracks. We focus on the 1st exam after online education and the *Gaokao*. For Chinese, online watching time had a positive effect on test scores for both exams, while both results are insignificant for the natural sciences track (columns 1 and 2 of panel A). For the social sciences, the coefficient for online watching time is positive in the 1st exam, but it is insignificant (columns 1 and 2 of panel B).

For math scores, online watching time has positive effects on math performance in the natural sciences, with 3.440 for the 1st exam and 2.061 for the *Gaokao* (columns 3 and 4 of panel A). This implies that the marginal effect of online education on math test scores is larger than that on Chinese test scores. Only in the 1st exam is the coefficient significant at the 5% level. This means that when the online watching for math increases by 10%, the score increased by approximately 0.344, holding other factors constant. However, the coefficients of online watching time are negative and insignificant in the social sciences (columns 3 and 4 of panel B).

Columns 5 and 6 of Table 2 present the regression results for English. The coefficient is positive for the 1st exam, but all coefficients are insignificant for both tracks. For comprehensive liberal arts scores (columns 7 and 8 of panel B), online education has a positive but insignificant effect on test scores. The coefficients are inconsistent for both exams for comprehensive science (columns 7 and 8 of panel A).

It is worth noting that previous scores have positive and significant effects on the 1st exam and the *Gaokao*, which suggests that the benefits of past efforts are not eliminated and learning is the result of sustained efforts. In addition, the 1st exam analysis of the natural sciences suggests that online education only has a positive and significant effect for math, while the effects are insignificant for other subjects (panel A). This may be due to the characteristics of different subjects. For subjects such as math, most of the class content is about problem-solving skills (Loria, 2014; Ramirez, Chang, Maloney, Levine, & Beilock, 2016). The longer the class time, the more proficient the students become at mastering their problem-solving skills, which suggests that the positive effect of online education will be more obvious. For language subjects such as Chinese and English, the effect of learning is the result of long-term accumulation of effort (Zhang, 2016; Zhang, 2017). Especially in the final year of high school, the content of those two courses is mostly based on

Table 2

The results of online education on the scores of the 1st exam and the Gaokao.

Variables	Chinese		Math		English		Comp. sci./lib. Arts	
	1st exam	Gaokao	1st exam	Gaokao	1st exam	Gaokao	1st exam	Gaokao
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Natural Sciences								
Ln(online edu. time)	0.264 (1.012)	0.469 (0.926)	3.440** (1.402)	2.061 (1.396)	1.598 (1.097)	−0.143 (0.728)	1.194 (2.044)	−1.942 (2.902)
Score before COVID-19	0.571*** (0.054)	0.684*** (0.064)	0.844*** (0.056)	0.759*** (0.029)	1.007*** (0.031)	0.828*** (0.040)	0.860*** (0.032)	0.758*** (0.052)
Observations	577	481	597	497	566	476	586	487
R-squared	0.241	0.269	0.496	0.447	0.789	0.582	0.573	0.46
Number of classes	19	17	19	17	19	17	19	17
Panel B: Social Sciences								
Ln(online edu. time)	0.854 (1.736)	−1.147 (0.749)	−2.172 (2.430)	−1.113 (1.840)	1.376 (1.705)	−0.098 (4.114)	1.089 (2.938)	0.172 (4.603)
Score before COVID-19	0.299 (0.165)	0.478** (0.091)	0.718*** (0.050)	0.730*** (0.035)	1.012*** (0.062)	0.777*** (0.087)	0.688*** (0.062)	0.652*** (0.064)
Observations	157	139	158	140	158	140	158	140
R-squared	0.118	0.281	0.482	0.574	0.823	0.728	0.582	0.557
Number of classes	4	4	4	4	4	4	4	4

Note: All regressions control for class fixed effects and individual and household characteristics. Robust standard errors are in parentheses, which are clustered at the class level. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

reviewing existing grammatical knowledge. Here, students' performance is mainly affected by their accumulation of knowledge in the past, so added class time may have little or no impact on their performance.

For social sciences students, the effects of online education on the two exams are insignificant (panel B). The results for liberal arts subjects are different from those for natural sciences subjects. When liberal arts students in China attend the *Gaokao*, their comprehensive subjects include geography, history, and political education, whereas the comprehensive subjects of natural sciences include chemistry, physics, and biology. In addition to what is gained from the knowledge in textbooks, the effectiveness of liberal arts learning requires a high level of extracurricular knowledge accumulation. The preceding analysis provides a possible explanation for the basic results. Also, the insignificant results may reflect the balance between online and offline effort. The students' online activity is to watch online courses, and the offline activity is to complete the homework assigned by teachers. Students' achievement is determined by the degree of participation in both online and offline activities. If students ignore offline activities, online education effect will also be offset, especially when teachers do not inspect the quality of homework, and even do not inspect whether or not students completed

Table 3

The heterogeneity analyses with respect to class classifications.

Variables	Chinese		Math		English		Comprehensive science	
	1st exam	Gaokao	1st exam	Gaokao	1st exam	Gaokao	1st exam	Gaokao
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Natural Sciences								
Ln(online edu. time)	1.659 (1.431)	1.671** (0.685)	12.770** (4.582)	0.162 (9.371)	3.002*** (0.603)	3.243*** (1.033)	2.502* (1.420)	−1.019 (6.531)
Ln(online edu. time)*	−2.166 (1.846)	−4.882* (2.542)	−9.715* (5.096)	1.077 (9.351)	−0.952 (1.301)	−2.467 (1.770)	0.637 (3.124)	0.516 (7.010)
second tire class	−0.535 (2.404)	−0.239 (1.766)	−8.625 (5.291)	5.429 (12.337)	−0.905 (2.633)	−5.439* (2.802)	0.926 (3.189)	14.659** (6.645)
Ln(online edu. time)*	−1.431 (2.186)	−0.369 (1.059)	−9.745* (4.887)	1.973 (9.446)	−2.152 (2.442)	−4.089** (1.632)	−3.789 (3.403)	−4.033 (7.641)
returning students	577	481	597	497	566	476	586	487
R-squared	0.242	0.276	0.498	0.448	0.789	0.583	0.574	0.465
Number of class	19	17	19	17	19	17	19	17
Panel B: Social Sciences								
Ln(online edu. time)	0.851 (1.121)	−0.08 (1.411)	−3.514 (4.186)	0.155 (1.311)	0.458 (2.426)	−5.187*** (0.616)	0.678 (2.749)	0.236 (2.482)
Ln(online edu. time)*	0.006 (2.754)	−2.389 (2.360)	2.952 (5.747)	−3.143 (2.311)	1.983 (2.664)	12.696*** (0.899)	0.358 (4.181)	0.915 (2.112)
returning students	157	139	158	140	158	140	158	140
R-squared	0.118	0.284	0.483	0.575	0.824	0.747	0.485	0.511
Number of classes	4	4	4	4	4	4	4	4

Note: All regressions control for class fixed effects. Robust standard errors are in parentheses, which are clustered at the class level. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

their homework.

In summary, the regression results show that online education has an obvious effect on math scores in the natural sciences but has no significant effect on students' test scores in the social sciences. In the following section, we add a series of interaction terms in the model to investigate whether the effects are different among various groups.

4.2. Heterogeneity analyses

The results above indicate that online education does have a positive effect on math scores in general, and it is possible that some groups may be affected more than others. For instance, some students could adjust their schedule immediately and adapt themselves to online education more quickly. Consequently, they can benefit more from online education. Thus, it is meaningful to investigate these heterogeneities.

4.2.1. Heterogeneity with respect to student classification

In this section, we interact tier classifications with online watching time to identify whether these effects differ with respect to class classification. As noted in the data section, the students in the natural sciences track are classified into four types: top-tier, second-tier, third-tier, and returning students. For the social sciences, there are two groups: new graduates and returning students. In the regressions, the top-tier and new graduates are the omitted groups for the tracks of natural sciences and social sciences, respectively. The results of these heterogeneity analyses are presented in Table 3.

Columns 1 and 2 report the results for Chinese. The top-tier students were the most positively affected by online education on the 1st exam and the *Gaokao*. For the social sciences, the results are insignificant for both exams. For math scores (columns 3, 4), the top-tier classes were still the most affected by online education, followed by third-tier, second-tier, and returning students in the 1st exam following online education in the natural sciences. With a 10% increase in online watching time, the math score increased by 1.277 in top-tier students, 0.306 in second-tier students, 0.415 in third-tier students, and 0.304 in returning students holding other factors constant. For English scores (columns 5, 6), the top-tier students were the most positively affected in natural sciences—the same as the results for Chinese—while in social sciences, returning students were the most positively affected. For comprehensive science (columns 7, 8 of panel A), third-tier students, rather than top-tier students, were the most positively affected tier. For comprehensive liberal arts

Table 4

The heterogeneity analyses with respect to household/individual characteristics.

	Natural sciences		Social sciences	
	1st exam	Gaokao	1st exam	Gaokao
Panel A: Chinese				
Variables	(1)	(2)	(3)	(4)
Ln(online edu. time)	0.44 (2.559)	−0.791 (1.335)	−1.257 (3.574)	−4.325 (7.809)
Ln(online edu. time)*teaching effectiveness (indifferent or better = 1)	0.633 (1.093)	0.593 (1.504)	7.933** (2.346)	−2.093 (2.922)
Panel B: Math				
Variables	(5)	(6)	(7)	(8)
Ln(online edu. time)	3.424 (3.007)	3.336 (2.698)	14.097* (5.731)	1.22 (9.588)
Ln(online edu. time)*device (computer or Tablet PC = 1)	−0.102 (2.542)	1.639 (2.466)	−10.8 (9.729)	−14.175* (5.742)
Panel C: English				
Variables	(9)	(10)	(11)	(12)
Ln(online edu. time)	−1.943 (2.311)	−3.543 (2.863)	1.645 (2.557)	9.346* (3.490)
Ln(online edu. time)*gender(male = 1)	3.011* (1.727)	−4.163 (2.457)	−3.244 (1.461)	−1.139 (0.958)
Ln(online edu. time)*study status at home (acceptable or fulfilling = 1)	0.635 (1.722)	8.238** (3.492)	3.135 (4.655)	−6.840* (2.702)
Panel D: Comp. science/Comp. liberal arts				
Variables	(13)	(14)	(15)	(16)
Ln(online edu. time)	−9.653** (3.973)	−0.006 (3.898)	8.910*** (0.951)	−1.001 (5.133)
Ln(online edu. time)*study status at home (acceptable or fulfilling = 1)	6.063* (3.317)	−0.942 (3.127)	−13.566* (4.614)	−7.018** (1.493)
Ln(online edu. time)*teaching Effectiveness (indifferent or better = 1)	5.579* (3.007)	−8.210** (3.745)	6.665 (5.611)	2.953 (10.605)

Note: All regressions control for class fixed effects. Robust standard errors are in parentheses, which are clustered at the class level. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The regression contains the same group of variables (interactions of urban, household consumption level, study status at home, and teaching effect) for every subject, and we report only the main results of each subject. The first two columns in each item are the results of the natural sciences, while the other two are the results of the social sciences.

(columns 7, 8 of panel B), returning students were the most positively affected by online education, but almost all coefficients are insignificant.

The above results come from eq. (1), with class classification interacting with online watching time, and imply that the top-tier students benefit most from online education. One possible explanation is that students in different tier classifications have variable adaptability to this new teaching method. Online education was proposed for all students to avoid delaying the progress of the course as the epidemic continued to become more serious, and most students had not been exposed to online classes before. It is possible that students with stronger adaptability are able to react better to extra stress, more flexibly adjust their time schedule, adapt sooner to the process, and better learn from online classes (Kuger, 2016; Martens & Lairmore, 2016). Thus, they naturally benefit more from online education.

4.2.2. Heterogeneity with respect to household and individual characteristics

In this section, we investigate whether online education effectiveness changes according to students' individual and household characteristics, which are viewed as crucial factors for students' learning (Blanden & Gregg, 2004; Caprara et al., 2008; Todd & Wolpin, 2003). Table B.1 summarizes the heterogeneity analyses in different respects alone. Based on these results, we select some important heterogeneity analyses results which involve gender, watching device, study status at home, and teaching effectiveness.

Table 4 shows the results. For teaching effectiveness, students who feel indifferent or better can benefit more in the social sciences of Chinese classes. However, the results are inconsistent for comprehensive science (columns 3, 13, 14). Students' subjective evaluation of courses also reflects their adaptability to online courses. Students who have better adaptability can gain more from online courses by actively adjusting their course arrangements.

For study at home, students whose self-evaluations are acceptable or fulfilling receive more benefits in online English education in the natural sciences track (columns 9, 10), and the coefficients of interaction are 0.635 and 8.238, respectively. For comprehensive science, the results are inconsistent, and the coefficients are 6.063 in the 1st exam and -0.942 in the *Gaokao* (columns 13, 14). Male students in the natural sciences obtained higher English scores on the 1st exam than their female counterparts, and the coefficient was 3.011 (column 9), while it was inconsistent for the *Gaokao* (column 10). For students who use computers or Tablets in online education, the coefficients are mostly negative but insignificant, implying that different tools may not affect online learning outcomes

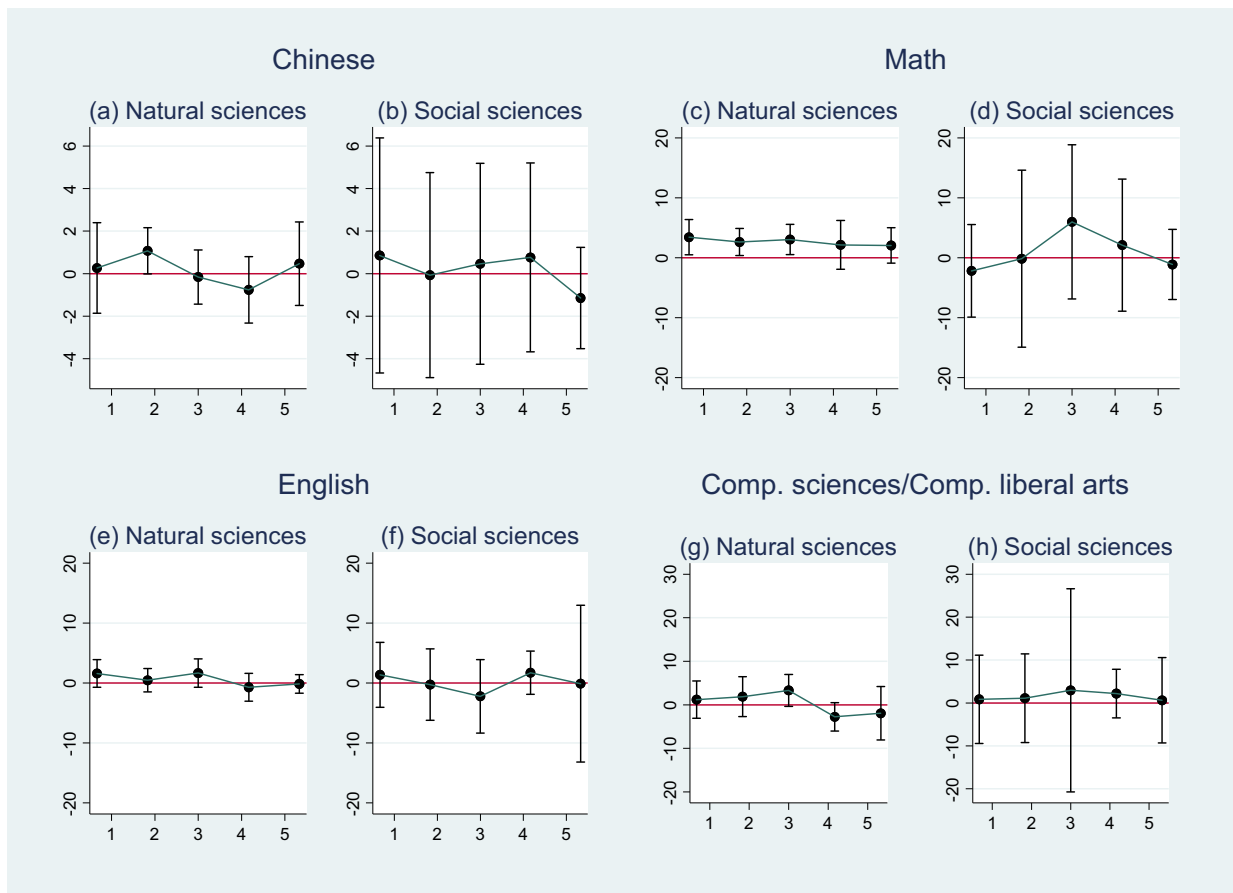


Fig. 4. The persistent impact of online courses on test scores.

Note: The vertical line is the 90% confidence interval. The number on the x-axis represents the time of exam taking after online education, e.g., '1' is the 1st exam taken after online education. In particular, the last exam is the national college entrance exam.

(columns 5–8). During the online learning period, most students only watch online courses and rarely interact with other students or teachers, so our article only pays attention to the watching time. That is to say in the online learning in our case, students did not make full use of advantages of computers and Tablets, though the different devices may have many different teaching effectiveness. This is because when the epidemic occurred suddenly, there is no sufficient time for teachers to get professional training. Nevertheless, we believe that the research on online teaching approaches is also very meaningful and we hope to make up for this in future research.

4.3. The persistence of the online education effect

Thus far, we have shed light on the effects for the 1st exam after online education and the *Gaokao*. The effects differ among subjects and tracks. In general, the effects are stronger and more significant for the 1st exam than for the *Gaokao*, which suggests that the online education effect may not persist long in regard to students' performance once offline education returns. To confirm this, we performed a regression analysis of scores for five exams taken after online education. Fig. 4 shows how the effect of online education changes over time. In each cell, the left side is the result in the natural sciences, and the right side is the result in the social sciences.

For the natural sciences, it is apparent that the effect is strong, positive, and steadily persistent over time for math. The online effect lasted until the first three exams for math, and the effect of math was between 0 and 10, with a 90% confidence interval. This suggests that the achievement gap in math during online education is not easy to overcome even when offline education returns. However, the online education effect is insignificant for other subjects in the five exams. For the social sciences, all coefficients are insignificant for all exams and all subjects.

The persistence of the online education effect varies greatly among subjects, which may be due to their characteristics. The online education effect of math is relatively small and only lasts until the third exam after returning to school, while the online education effects of other subjects are insignificant and do not persist for long. In the teaching process of math, teachers emphasis more on skills and cultivate students' logic, which will firstly shape their thinking habits and then affect their whole learning process (Franke & Kazemi, 2001; Huang, Ricci, & Mnatsakanian, 2015). Math has a strong logic component, and the knowledge involved is interdependent, whereby the learning gap caused by online courses is not easy to eliminate. Chinese and English are subjects that emphasize the use of language. Teachers in the third year of high school usually summarize the topics and skills in lectures, and the learning of language mainly depends on comprehensive accumulation of knowledge, so teachers play a limited role in guiding students in the classroom (Zhang, 2016; Zhang, 2017). The results of comprehensive science and liberal arts are both insignificant and do not persist for long. This may be because these two subjects involve a wide range of knowledge. Each subject is composed of three smaller subjects. It is therefore difficult to improve test scores in a short time during online education.

4.4. The distributional effects of online education on test scores

We have demonstrated the average effects of online education on students' test scores and how the average effects differ across a few factors. However, the effects may also vary in relation to student performances. To investigate this source of variation, we estimated the effects of online education time in different quantiles of test scores using recentered influence function (RIF) regression (Firpo, Fortin, & Lemieux, 2009).¹³

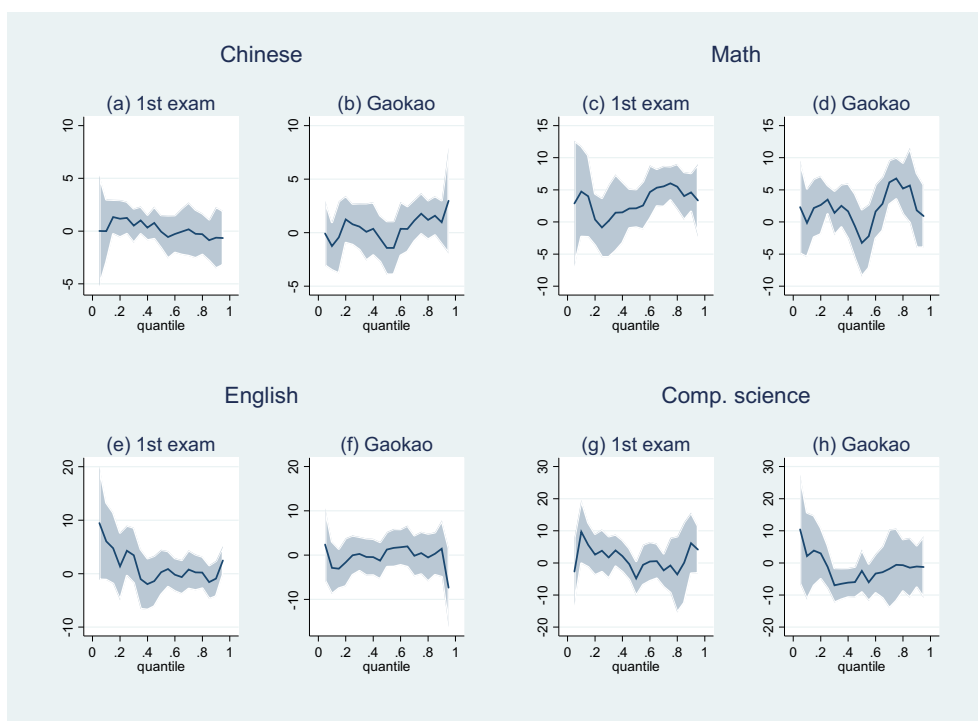
Fig. 5(a) presents the results of natural sciences. The positive effect of online education of math on performance is obvious for the 0.60th to 0.80th quantiles in the 1st exam and the *Gaokao* after online education. For Chinese and English, the effects are insignificant in both exams. In comprehensive science, the effect is insignificant in the 1st exam but surprisingly negative for the 0.30th to 0.50th quantiles. Fig. 5(b) shows the results of social sciences. For Chinese and comprehensive liberal arts, there are negative effects in some quantiles.

In general, online education does affect students' academic achievement, but the positive effect is limited for students in the 0.60th to 0.80th quantiles of math performance in the natural sciences, which can only improve their scores by more than 0.25 with a 10% increase in watching time in the 1st exam and the *Gaokao*. Surprisingly, we also found significant negative effects of online education in some subjects. Of course, the negative results may also be due to differences in teaching quality, teachers' ability to convert offline content into online content and other factors among classes. After all, the epidemic happened very suddenly. There is no sufficient time for teachers to get professional training and no sufficient time for students to get used to online education. Consequently, the benefits of online learning have not been fully achieved.

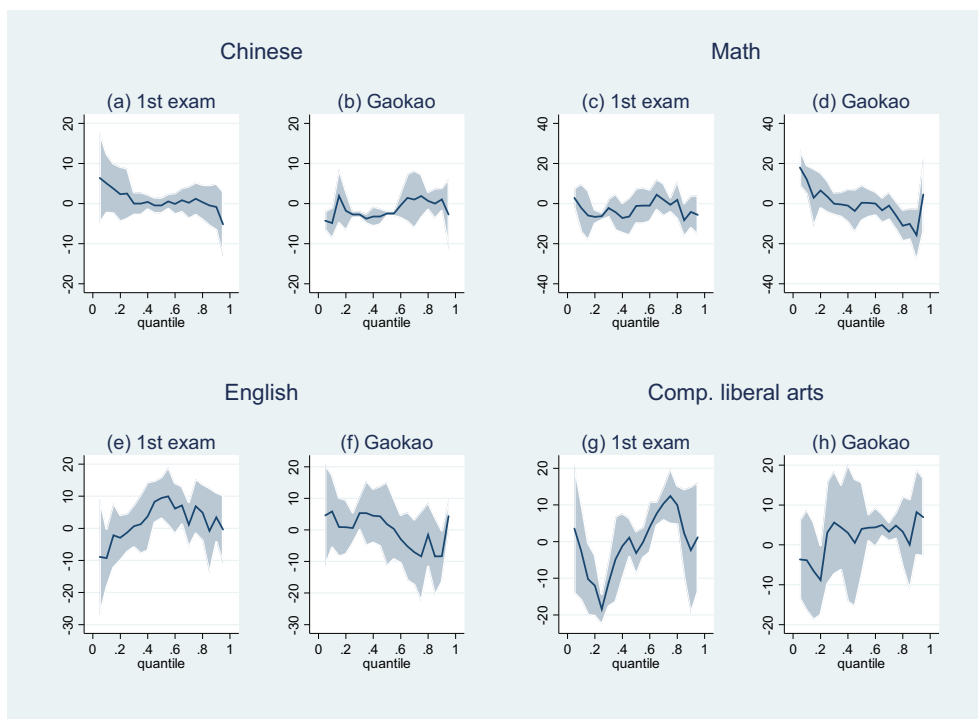
Recently, people have been concerned about whether COVID-19 is widening the educational learning gap (Anderson, 2020). Luo and Si (2020) found that students from low-income families and rural families had a relatively low performance compared with their grade average before COVID-19. They suggest that online education has expanded the learning gap between different family conditions. As the above analysis shows, online education positively affects academic achievement for students in the 0.60th to 0.80th quantiles of math in the natural sciences. It is an open question as to whether online education widens the inequality of academic performance, so we estimate the effects of online education on inequality measured by variance and the Gini coefficient in RIF regression.

Table 5 presents the regression results for the inequality of the test scores in the 1st examination after returning to school and the *Gaokao*. In each cell, the left side shows the variance, and the right side shows the Gini coefficient. The coefficients are all insignificant,

¹³ See Appendix C for a brief introduction to RIF regression.



(a) Natural sciences



(b) Social Sciences

Fig. 5. The distributional effects of online education on test scores.

Note: The shadow area is the 90% confidence interval.

Table 5

The inequality analysis in the 1st test after returning to school and the Gaokao.

Variables	Nature sciences				Social sciences			
	1st exam		Gaokao		1st exam		Gaokao	
	Variance	Gini	Variance	Gini	Variance	Gini	Variance	Gini
Panel A: Chinese								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ln(online edu. time)	−22.379 (19.870)	−0.006 (0.005)	9.618 (11.710)	0.003 (0.004)	−14.475 (22.396)	−0.007 (0.011)	23.634 (28.768)	0.009 (0.011)
Panel B: Math								
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Ln(online edu. time)	10.593 (51.679)	−0.001 (0.010)	23.658 (46.788)	0.002 (0.009)	5.047 (81.310)	0.008 (0.018)	−111.585 (83.018)	−0.025 (0.021)
Panel C: English								
	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)
Ln(online edu. time)	−92.639 (80.868)	−0.016 (0.012)	−27.836 (65.842)	−0.003 (0.012)	105.615 (54.441)	0.016 (0.008)	−103.832 (85.665)	−0.023 (0.020)
Panel D: Comp. science/Comp. liberal arts								
	(25)	(26)	(27)	(28)	(29)	(30)	(31)	(32)
Ln(online edu. time)	−133.435 (123.294)	−0.011 (0.008)	54.957 (243.031)	0.003 (0.013)	12.846 (259.309)	0.008 (0.015)	150.168 (90.305)	0.01 (0.007)

Note: Robust standard errors are in parentheses, which are clustered at the class level. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. In each cell, the left side is about variance and the right side is about Gini index. The first four columns in each item are the results of natural sciences, while the others are the results of social sciences. A coefficient of 0 in this table does not mean that regression result is 0 since we retain 3 digits after the decimal point. For example, if the result is 0.0001 then the coefficient will be 0 in this table.

which suggests that online education has not changed the distribution of students' achievement. This result may be due to the short duration of the online courses, which was less than two months. In addition, the key elements of online education are proper instructional design and training teachers in the relevant technological skills (Carlana & Ferrara, 2021; Ke, 2010). During online education, teachers did not receive relevant training because the epidemic started very suddenly. Moreover, network connection problems greatly reduced the effectiveness of online education.

5. Conclusion

Since online education is playing an increasingly important role in the modern education system, it is necessary to correctly estimate its impact on student learning outcomes. In this paper, the basic model includes students' previous scores from exactly before the start of online education, which is a sufficient statistic of their unobserved ability and motivation, to try to adjust for endogeneity problems. We found that, on average, online education has positive yet moderate impacts on test scores, especially for math in natural sciences. Then, interactions were added to the basic model to estimate whether the effects differed among student tier classifications and demographic characteristics. The top-tier students were the most positively affected by online education. For students of different backgrounds, there were also great differences in online education effect. For instance, for students who use computers or tablets in online education, the coefficients are insignificant, which implies that different tools may not affect the benefits of online learning. Next, results of five exams that followed the completion of online education were used as independent variables to explore whether the effects lasted. The results show that the online education effect only persisted for math for the natural sciences in general. Finally, distributional effects using quantile regression and inequality effects using variance and the Gini index were discussed. The distributional effects show that students in the 0.60th to 0.80th quantiles of math can only increase their scores by more than 0.25 with a 10% increase in online watching time. In the analysis of inequality, it was found that online education neither changes the distribution of student performance nor widens inequality.

Other subjects did not demonstrate a positive significant effect of online courses, but we could not clearly identify whether the results were due to online teaching or the actual teaching format, given that there was no control group in our article. Of course, these results may directly result from the characteristics of certain subjects. For instance, English scores have a lot to do with accumulated knowledge, so the hours of lecture during Grade Three of high school may not have a significant impact on them. However, the quantile regression results for liberal arts suggest that longer watching time can significantly reduce test score. The possible explanation may be due to differences in teaching quality among classes, teachers' ability to convert offline into online content, and many other factors.

Although online education provides opportunity to real-time interaction, the online education in our case does not make full use of its advantages. Thus, our results may underestimate the effect of online education. Successful online education requires proper instructional design and teaching elements (Ke, 2010). During the period of online education we evaluated, teachers did not receive professional training because the epidemic happened suddenly. Unskilled use of online education equipment, an unreasonable curriculum, and poor internet conditions greatly reduced the effectiveness of some classes. The time arrangement of these online courses was the same as offline learning. For students of different grades, an online schedule should be set according to the specific situation of the student and the course and should not be a one-size-fits-all, which has a negative impact on the online education effect. Finally,

students were uncertain about the duration of online classes, and this expectation also affected the effectiveness of online education.

Education has always been an important topic and a primary approach to promote social equity. In addition, the human capital developed by education is very important to a country's economic development (Grossman, 2000; Nelson & Phelps, 1966). Education can improve the quality, ability, and technical level of the labor force, thereby increasing labor productivity. Time spent on education, as a scarce resource, is critical to the educational achievement. The allocation of this time should not be constrained by political goals but should be based on how it increases students' motivation. Online classes provide us with a new perspective concerning teaching time, as students choose their own learning time during online classes. This is the first large-scale use of online courses as the main education method in human history. We look forward to online education playing a more important role in the future.

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Appendix A. The correlation between the 1st exam test score and online education time

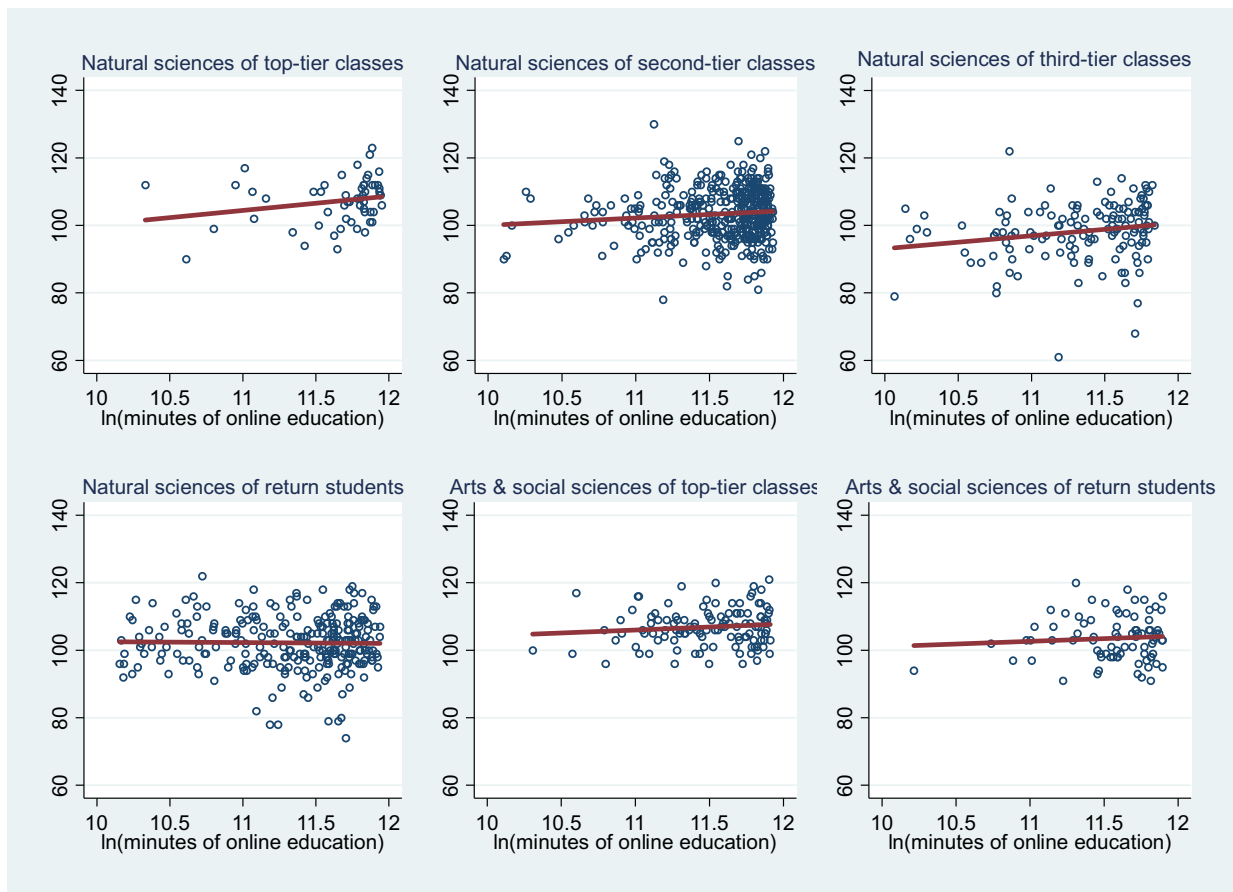


Fig. A.1. Chinese test score vs. online education time (by classifications & tracks).

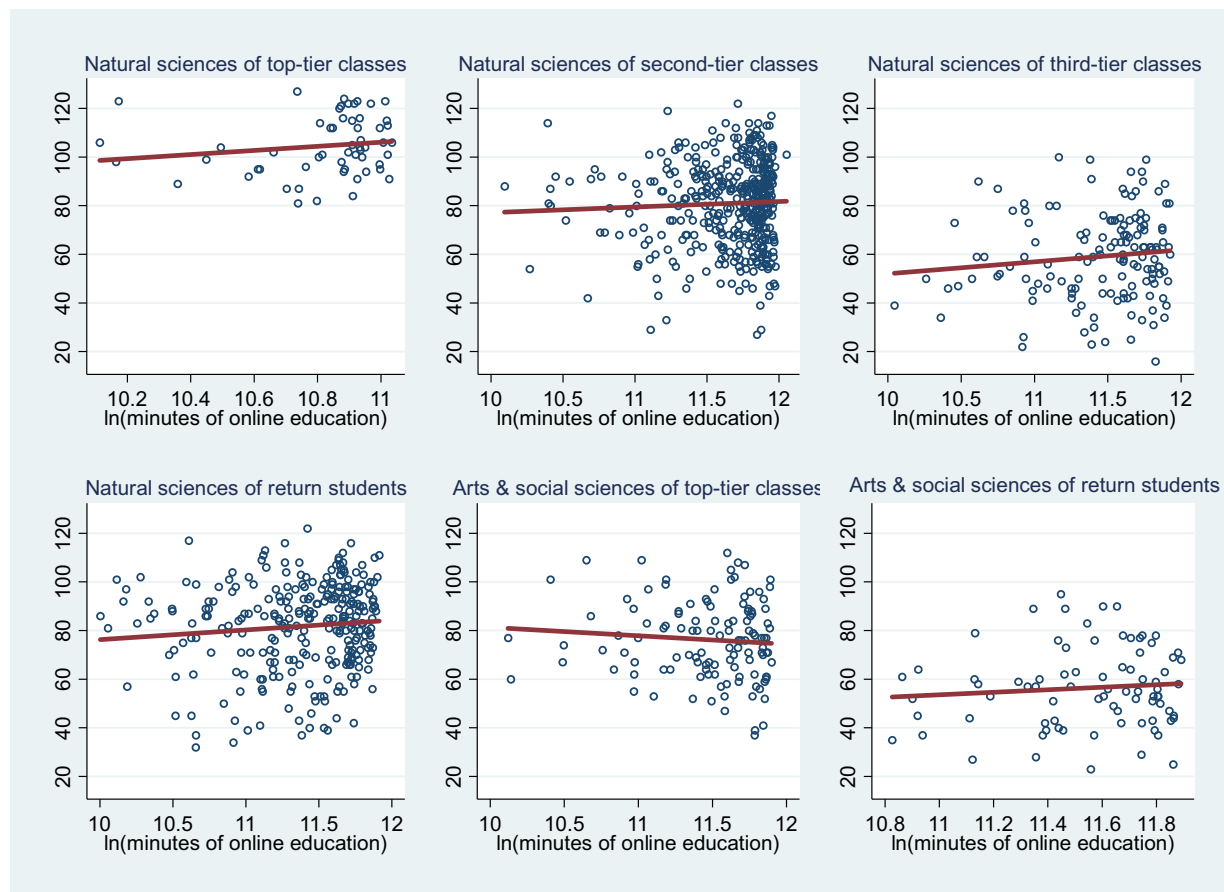


Fig. A.2. Math test score vs. online education time (by classifications & tracks).

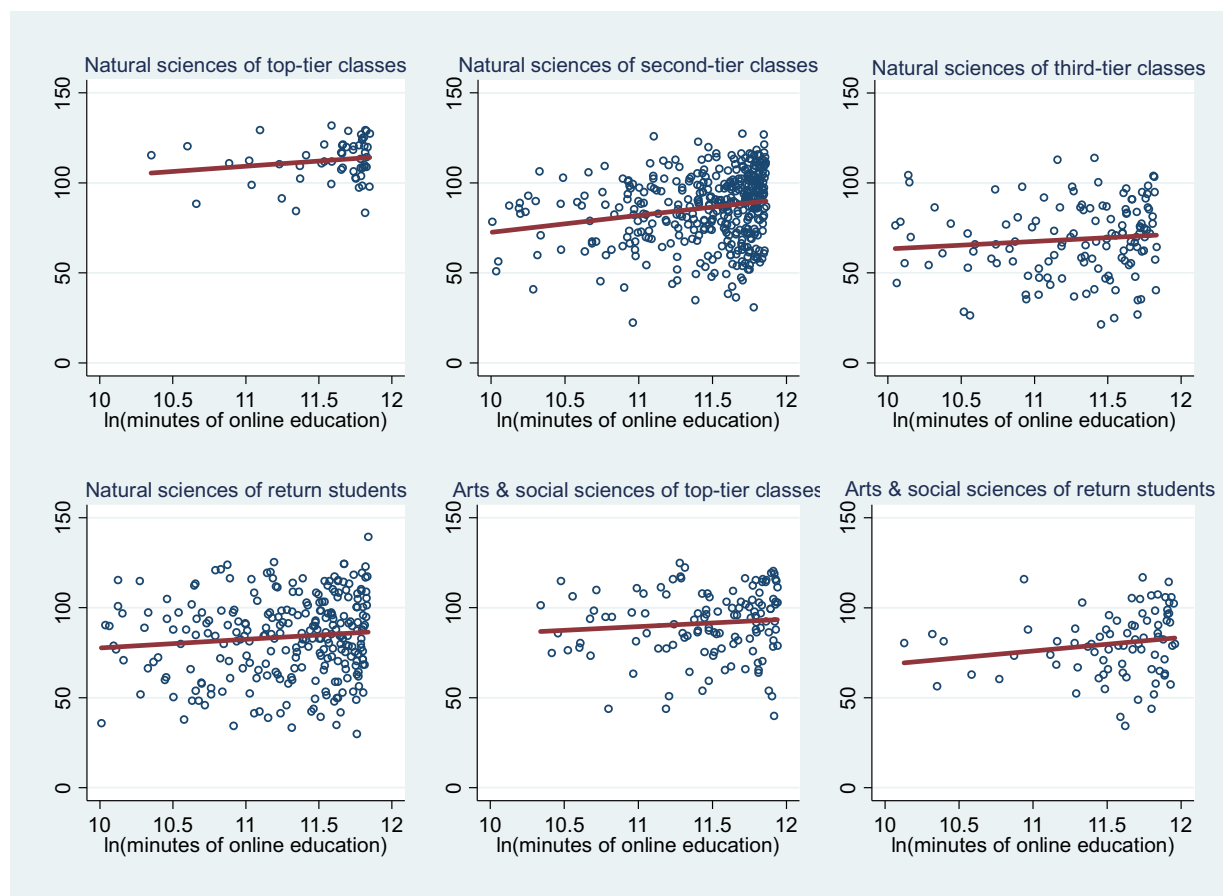


Fig. A.3. English test score vs. online education time (by classifications & tracks).

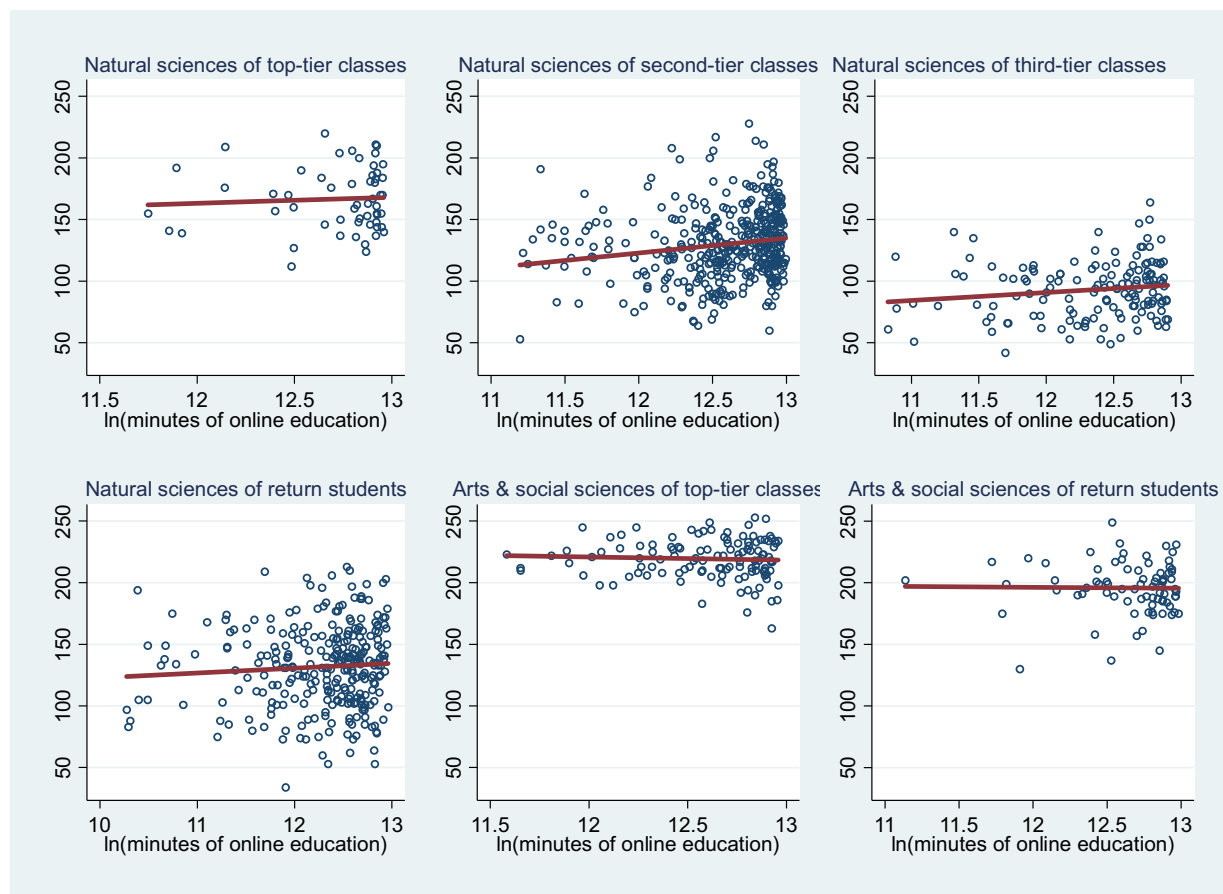


Fig. A.4. Comprehensive science and art test score vs. online education time (by classifications & tracks).

Appendix B. Heterogeneity analyses

Table B.1
Heterogeneity analyses.

	Chinese		Math		English		Comprehensive science/ liberal arts	
	Nature science	Arts/social sciences	Nature science	Arts/social sciences	Nature science	Arts/social sciences	Nature science	Arts/social sciences
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Father's education*ln (online education time)		5(+)	2(+),3(+)					
Mother's education*ln (online education time)	5(+)		3(+)		2(-)			5(+)
Household consumption level*ln (online education time)	2(+)							
Parental supervision*ln (online education time)	3(-)	1(+)				1(-)		
Watching device*ln (online education time)	2(+)			5(-)			3(+)	
Everyday learning time*ln (online education time)							4(-)	
Whether attend class on time*ln (online education time)	5(-)							2(-),3(-)
Attitude in online courses*ln (online education time)	2(+)			2(-),5(+)		5(+)		
Prefer online or traditional*ln (online education time)					3(+),4 (+),5(+)		3(+)	
Study status at home*ln (online education time)		3(+),4(+)		1(-),2(-)	5(+)		1(+)	1(-),2(-)
Teaching effect*ln (online education time)		1(+)					1(+),2(+)	
Gender*ln (online education time)	3(-)	1(-)				1(-)	2(+)	
Urban*ln (online education time)	2(+)		1(-),3(+)	4(+)	2(-)		2(+)	5(+)

Note: All regressions control only one interaction. '1' means that the dependent variable is the 1st exam after online education. '2' means that the dependent variable is the 2nd exam after online education. '3' means that the dependent variable is the 3rd exam after online education. '4' means that the dependent variable is the 4th exam after online education. '5' means that the dependent variable is the *Gaokao*. '+' means that the interaction is positive and significant at the 1%, 5%, or 10% level, while '-' means it is negative and significant at the 1%, 5%, or 10% level. For instance, 5(+) means that when the dependent variable is *Gaokao*, this interaction is positively significant.

Appendix C. RIF regression

The re-centered influence function (RIF) is defined as

$$\text{RIF}(y; v(F_Y), F_Y) = v(F_Y) + \text{IF}(y; v(F_Y), F_Y), \quad (\text{C.1})$$

where $v(F_Y)$ is a distributional statistic of interest, and $\text{IF}(y; v(F_Y), F_Y)$ is the influence function of this specific distributional statistic. The influence function $\text{IF}(y; v(F_Y), F_Y)$ of the $v(F_Y)$ represents the influence of an individual observation on that statistic. Adding back the statistic $v(F_Y)$ to the influence function yields the so called re-centered influence function (RIF). A convention feature of the RIF is that its expectation is equal to $v(F_Y)$. One of the most important strengths of the above expression is that it "can be used directly for the estimation of standard errors of any statistic for which a RIF exists (Rios-Avila, 2020)".

The RIF regression uses the estimated $\text{RIF}(y; v(F_Y), F_Y)$ for each observation y_i in the data as the dependent variable and regress it against all the variables of interest:

$$\text{RIF}(y; v(F_Y), F_Y) = X \cdot \varphi + \varepsilon, E(\varepsilon) = 0. \quad (\text{C.2})$$

when $v(F_Y)$ is the quantile, the RIF regression becomes the unconditional quantile regression (Firpo et al., 2009). The coefficient φ gauges the impact of the change in the distribution of an explanatory variable on the marginal distribution of Y .

Appendix D. DINGDING

DINGDING is a virtual meeting platform. It was used as a live broadcast platform by schools for online classes during COVID-19. On DINGDING, the school established class groups and determined teachers in charge of the class, so that students and parents will also join the group. When the teacher is in class, he/she initiates a live broadcast through DINGDING, then students can watch the live

broadcast in the specific time. DINGDING has students' total watching time information for every student. When you turn off the live website and then come back, the time is accumulated. Students can replay the videos after the live class. The watching time is the sum of the time you watched before and you watched repeatedly. Fig. D.1 shows a screenshot of DINGDING.



Fig. D.1. The interface of DINGDING.

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